usage with Savings Plans. Reducing costs is also possible by committing to a smaller fraction of your workload.

3.8.2 Taking advantage of spare compute capacity

AWS is operating data centers at large scale, which results in spare capacity because it has to build and provision data centers and machines in advance to be able to fulfill future needs for on-demand capacity. But spare capacity does not generate revenue. That's why AWS tries to reduce spare capacity within its data centers. One way of doing so is offering Spot Instances.

Here is the deal. With Spot Instances, you get a significant discount on the on-demand price without the need to commit to using capacity in advance. In turn, a Spot Instance will start only when AWS decides that there is enough spare capacity available. In addition, a Spot Instance might be terminated by AWS at any time on short notice.

For example, when writing this on April 8, 2022, the price for an `m5.large` Spot Instance in US East (N. Virginia) is $0.039 per hour. That's a discount of about 60% compared to the $0.096 on-demand price. The spot price for EC2 instances used to be very volatile. Nowadays, it changes much more slowly.

But who is crazy enough to use virtual machines that might be terminated by AWS at any time with notice of only two minutes before the machine gets interrupted? Here are a few scenarios:

- Scanning objects stored on S3 for viruses and malware, by processing tasks stored in a queue
- Converting media files into different formats, where the process orchestrator will restart failed jobs automatically